

DOCUMENT 001

ATAI 8WD – Multi-Mechanism Layer Separation Architecture

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Program: ATAI 8WD Hybrid Tri-Layer Aerospace Platform

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1. Executive Overview

This document defines the certified separation architecture between Layer-1, Layer-2, and Layer-3 of the ATAI 8WD platform. The system is designed under the principle:

Maximum Operational Safety with Minimum Mechanical Complexity

The architecture incorporates:

- Multi-stage autonomous separation logic
- Dual-actuator structural release
- Dual-person mechanical reset override (ground only)
- Redundant sensing and load validation
- Dedicated access corridors integrated into structural design

The system enables safe, repeatable, and inspectable separation of layers under controlled operational conditions.

2. Architectural Philosophy

The ATAI 8WD tri-layer platform consists of:

- Layer~~1~~¹: Primary lift and propulsion platform
- Layer~~2~~²: Intermediate structural and systems layer
- Layer~~3~~³: Independent flight capsule

Layer separation is treated as a safety-critical aerospace event comparable to stage separation in launch vehicles.

Design philosophy includes:

1. Separation must be computer-sequenced but pilot-authorized.
2. Structural release must never occur under unsafe load conditions.
3. Manual mechanical override must exist for ground reset.
4. Override must require dual-person synchronized action.
5. No manual override is permitted during flight.

3. Multi-Mechanism Separation System (MMSS)

3.1 System Definition

The Multi-Mechanism Separation System (MMSS) is composed of:

- Four Primary Structural Hardpoints (PSH)
- Dual Redundant Electro-Hydraulic Actuators per hardpoint
- Independent Mechanical Reset Shaft
- Dual-Key Interlocked Override Interface
- Load Sensors
- Position Sensors (Locked/Unlocked)
- Separation Gap Sensors

Each interface (L1-L2 and L2-L3) includes four symmetric PSH units arranged at 90° intervals.

4. Three-Stage Separation Sequence

Stage 0 – Arm and Validation

- Independent flight computers active (Layer-3)
- Capsule propulsion spooled
- Cross-layer power transfer complete
- Load window verification
- CG alignment confirmation
- Pilot authorization

Separation cannot proceed unless all green conditions are satisfied.

Stage 1 – Utility Disconnect

All non-structural interfaces disconnect first:

- Electrical power
- Data buses
- Hydraulic lines
- Fuel lines

Quick-disconnect, self-sealing blind-mate connectors are used.

Failure response: - Retry attempt - Secondary actuator activation - Localized isolation or controlled severance (ground only)

Stage 2 – Structural Release

Each Primary Structural Hardpoint performs:

- Primary actuator unlock command

- Secondary actuator verification
- Locked sensor cross-check
- Load drop confirmation

Structural release is only permitted if:

$$\text{Load} < \text{Certified Release Threshold}$$

Latch geometry is tapered and self-aligning to prevent shear binding.

Stage 3 – Positive Separation

After unlock confirmation:

- Capsule lift fans increase thrust
- Lateral micro-thrusters stabilize orientation
- Gas-assisted pushers create separation impulse
- Separation gap sensors verify clearance

System transitions to Independent Flight Mode.

5. Dual Person Mechanical Reset Architecture

5.1 Purpose

To allow safe ground reset of jammed or partially released latches. The system is NOT designed to force release under load. It is designed to:

- Reset actuator indexing
- Clear mechanical cam misalignment
- Re-establish known locked state

5.2 Corridor Access Integration

Four structural access corridors are integrated into Layer-1 and Layer-2. Each corridor provides:

- Protected access to one PSH unit
- Mechanical override port
- Physical lock position indicator
- Maintenance lighting and safety anchoring points

Corridors are reinforced load-neutral structural tunnels that do not transfer primary flight loads.

5.3 Dual-Key Interlock Mechanism

Each PSH includes:

- Two separated mechanical key ports
- Geared synchronization shaft
- Torque-balanced cam reset system

Mechanical reset requires:

1. Two operators positioned at opposing access points
2. Simultaneous key insertion
3. Coordinated rotational input within defined tolerance window
4. Mechanical confirmation flag transition

Single-person override is mechanically impossible.

6. Ground-Only Activation Safeguards

Manual reset is physically blocked unless:

- Thrust = 0
- Platform weight fully supported by landing structure
- Load sensors confirm minimal structural tension

- Maintenance mode electronically enabled

A mechanical safety gate prevents key insertion in flight.

7. Hardpoint Mechanical Design

Each PSH contains:

- Oversized load-bearing latch
- Tapered engagement cone
- Dual release cams
- Shear-protected actuator drive
- Independent reset shaft
- Visual lock status window

The reset shaft does not directly move the latch under load; it re-indexes the release cam assembly only when structural loads are neutralized.

8. Failure Mode Handling

The MMSS accounts for:

- Single latch jam
- Asymmetric release
- False unlocked indication
- Partial separation
- Utility disconnect failure

Recovery hierarchy:

1. Automated retry
 2. Load relief maneuver
 3. Secondary actuator command
 4. Abort and maintain attached flight
 5. Ground mechanical reset
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9. Corridor Convergence Nodes

At the junction of the three layers:

- Corridor pathways terminate at Structural Interface Nodes (SIN)
- SIN modules house PSH assemblies
- Utility routing is segregated from structural latch housing
- Thermal isolation is integrated
- Shock-absorbing mounting frame used

The geometric intersection of corridors and SIN ensures:

- Direct maintenance access
- Visual inspection capability
- Rapid mechanical intervention
- Minimal structural disturbance

10. Operational Safety Logic

Separation is:

Pilot Authorized

Computer Sequenced

Sensor Validated

Physics Confirmed

No structural release occurs based solely on command signal. Release confirmation requires:

- Position sensor agreement
 - Load drop detection
 - Gap formation detection
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11. Certification Intent

The Multi-Mechanism Separation System is designed to meet future aerospace certification criteria for:

- Redundancy
 - Fail-safe operation
 - Maintainability
 - Human-factor safety
 - Ground serviceability
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12. Engineering Conclusion

The ATAI 8WD Multi-Mechanism Separation Architecture provides:

- Controlled three-stage autonomous separation
- Redundant structural release
- Dual-person mechanical reset capability
- Corridor-integrated maintenance access
- Ground-only override safeguards

This system enables Generation-1 deployment with high operational confidence while preserving future scalability for larger platform variants.

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